Lecture 21 Notes

Recap: Matrix Norms Leading to SVD

The instructor recaps the matrix-norm results from L20:

- One way to define matrix norms is to **vectorize** A and apply a familiar vector p-norm. This treats the matrix as a vector in $\mathbb{C}^{mn}$, not as a linear map.
- **Frobenius norm** (flatten then 2-norm): $\|A\|_F = \left(\sum_{i,j} |a_{ij}|^2\right)^{1/2}$ — sums the energies of all entries; does **not** care where the entries sit (no linear-map perspective).
- **Induced norms** treat $A : \mathbb{C}^n \to \mathbb{C}^m$ as a mapping; $\|A\|_{q,p} = \sup_{x \neq 0} \|Ax\|_q / \|x\|_p = \sup_{\|x\|_p = 1} \|Ax\|_q$.
  - $\|A\|_{1,1} =$ max column ℓ_1-norm.
  - $\|A\|_{\infty,\infty} =$ max row ℓ_1-norm.
  - $\|A\|_{2,2} = \sqrt{\lambda_{\max}(A^*A)} =$ length of the **longest principal semi-axis** of the ellipsoid that is the image of the unit ℓ_2-ball.

That last picture — **unit sphere maps to an ellipsoid** — is the geometric center of the **Singular Value Decomposition (SVD)**. This lecture presents the SVD from an **algebraic** view and a **geometric** view, then gives a **formal existence proof**.

SVD: The Algebraic View

The SVD states that **any** matrix A (rectangular allowed) can be written

$$A = U\Sigma V^*,$$

where: - U is $m \times m$ **unitary** (left singular vectors), - V is $n \times n$ **unitary** (right singular vectors), - Σ is $m \times n$ **rectangular diagonal** (the (i,i) entries may be nonzero, all others zero), with **real, non-negative** diagonal entries $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.

The relative sizes follow the shape of A: if A is **fat**, V is the larger unitary; if A is **tall**, U is larger. This fits the course storyline of writing A as a product of **simple matrices** — here **two unitaries and one (real, nonneg) diagonal**.

On the conjugate transpose V^*. It is a convention to write the third factor as V^* rather than V: multiplying by V^* means taking inner products with the right singular vectors. (Geometrically V might read more naturally, but the $U\Sigma V^*$ convention is standard.)

Connection to (and Departure from) Similarity

Recall the **similarity** story for a **square** $A : \mathbb{C}^n \to \mathbb{C}^n$: pick **one** basis T for **both** input and output; the map's new representation is $T^{-1}AT$; we asked whether we can choose T to make this **diagonal**. Answer: **not for all** matrices (only diagonalizable ones). The restriction was using the **same basis** for input and output.

The key step to SVD: drop that restriction. Use a basis T_1 for the input and a **different** basis T_2 for the output; the representation becomes $T_2^{-1}AT_1$. Question: can we always find T_1, T_2 making this **diagonal**, even for **rectangular** A (where input and output dimensions differ, so the **same** basis is impossible)? **Answer: yes — always — using orthonormal bases.** That is exactly the SVD: choosing one orthonormal basis for the input (V) and one for the output (U) renders **any** linear map diagonal, with real nonneg diagonal entries. "I can represent any linear mapping by a diagonal matrix if I choose my coordinate axes wisely — one set for the input, one for the output."

SVD: The Geometric View

Image of a Sphere Is an Ellipsoid

Two facts about linear maps: 1. The image of a **line segment** is a line segment (shown earlier for the ℓ_1-ball). 2. The image of an **ellipsoid** (in particular the unit sphere) under a linear map is an **ellipsoid** — shape preserved.

A **line segment is a degenerate ellipsoid** (one semi-axis has length zero). So degenerate cases fit the same picture: - A **rank-1** 2×2 map sends the circle to a line segment (an ellipse with second semi-axis = 0). - A map from 3-D to 2-D sends a sphere to an ellipse in the plane; if the matrix is not full rank, that ellipse can collapse to a line segment. - A **tall** 3×2 map (rank ≤ 2) sends a 2-D circle to an ellipse lying in a plane through the origin (the range space). If the rank is only 1, the image is again a line segment.

The instructor notes that this geometric discussion is based on the hand-drawn/pre-lecture figure: a sphere/circle maps to an ellipsoid/ellipse, with degenerate cases handled by allowing zero semi-axis lengths.

Student Q&A: What Is an Ellipsoid?

A student asks for the definition. An ellipsoid is the **level set of a quadratic function**:

$$\{x : (x - x_0)^* A(x - x_0) \leq b\},$$

with A **positive definite**. (A positive **semi**definite A gives a **degenerate** ellipsoid — some dimensions collapse.) Geometrically you "cut" the quadratic at a level and look at the region in the domain.

Assigning Singular Values and Vectors

Accept (formal proof later) that the image of the unit sphere is an ellipsoid. Label its components: - The **first (longest) principal semi-axis** has length σ_1 along the unit vector u_1: the axis is $\sigma_1 u_1$. Its **pre-image** on the input sphere is the unit vector v_1. So $Av_1 = \sigma_1 u_1$. - The **second** semi-axis is $\sigma_2 u_2$, with pre-image v_2: $Av_2 = \sigma_2 u_2$. - For a rank-r matrix there are r nonzero semi-axes $\sigma_1 \geq \cdots \geq \sigma_r > 0$ with $Av_k = \sigma_k u_k$, $k = 1, \ldots, r$.

Student check: why does such a v_k exist? Because $\sigma_k u_k$ is chosen as a point on the image of the unit sphere under A; by definition of image, at least one unit input vector maps to that point.

The u_k (semi-axis directions in the **output** space) are **orthogonal by definition** of principal axes. The pre-images v_k (in the **input** space) are also **orthonormal** — not obvious, proved formally below.

Matrix Form (Reduced)

Collecting the pre-images as columns of $\hat{V} = [v_1 \ \cdots \ v_r]$ ($n \times r$) and the axis directions as $\hat{U} = [u_1 \ \cdots \ u_r]$ ($m \times r$):

$$A\hat{V} = \hat{U}\hat{\Sigma}, \qquad \hat{\Sigma} = \operatorname{diag}(\sigma_1, \ldots, \sigma_r) \ (r \times r, \text{ positive diagonal}).$$

This relation records the nonzero action of A: the v_k's generate the principal semi-axes. After the null-space directions are added and the zero actions are accounted for, the zero terms can be removed again to give the **reduced (compact) SVD**

$$A = \hat{U}\hat{\Sigma}\hat{V}^*.$$

Extending to the Full SVD and the Four Fundamental Subspaces

$\hat{V}$ and $\hat{U}$ have only r orthonormal columns (orthonormal, but not unitary). Extend: - Add $n - r$ orthonormal columns to $\hat{V} \to$ unitary V. The added columns are arbitrary subject to completing an orthonormal basis, and in this construction they span the null space: $Av_k = 0$ for $k > r$ (they are silenced by the zero columns/diagonal positions of Σ). - Add $m - r$ orthonormal columns to $\hat{U} \to$ unitary U. These added columns are also arbitrary subject to completing U to unitary; they do not contribute to A because their coefficients in Σ are zero.

The $r \times r$ positive-diagonal block thus extends to the $m \times n$ rectangular diagonal Σ (first r diagonal entries nonzero, rest zero). The geometric construction first gives

$$AV = U\Sigma.$$

Multiplying on the right by V^* gives the **full SVD** $A = U\Sigma V^*$. The columns immediately give **orthonormal bases for all four fundamental subspaces**:

Subspace	Basis	SVD columns
Row space $\mathcal{R}(A^*)$	$v_1, \ldots, v_r$	first r cols of V (nonzero outputs)
Null space $\mathcal{N}(A)$	$v_{r+1}, \ldots, v_n$	last $n - r$ cols of V ($Av_k = 0$)
Column space $\mathcal{R}(A)$	$u_1, \ldots, u_r$	first r cols of U
Left null space $\mathcal{N}(A^*)$	$u_{r+1}, \ldots, u_m$	last $m - r$ cols of U

So the SVD hands us the **rank** (number of nonzero σ_k) and orthonormal bases for all four subspaces at once — "an excellent analysis tool."

Outer-Product (Rank-1 Sum) Form

Expanding $U\Sigma V^*$:

$$A = \sum_{k=1}^{r} \sigma_k \, u_k v_k^*.$$

The multiplication is: $\hat{U}\hat{\Sigma} = [\sigma_1 u_1 \;\; \cdots \;\; \sigma_r u_r]$, while $\hat{V}^*$ has rows $v_1^*, \ldots, v_r^*$; multiplying column-by-row gives the sum of outer products.

Each $\sigma_k u_k v_k^*$ is **rank one** (column vector $\times$ row vector). Summing r of them gives a rank-r matrix. Ordered by $\sigma_1 \geq \sigma_2 \geq \cdots$, the first term is the "most important." This is the form you see in many papers; it is an **orthonormal expansion** of A in a basis **adapted to A itself** (derived from A).

The Matrix Inner Product and Orthogonality of Rank-1 Components

The rank-1 terms $\sigma_k u_k v_k^*$ are **orthogonal** with respect to the **matrix inner product** — the extension of the Euclidean inner product to matrices:

$$\langle A, B \rangle = \operatorname{tr}(B^* A).$$

Proof of orthogonality. For $i \neq j$,

$$\left\langle u_i v_i^*, \, u_j v_j^* \right\rangle = \operatorname{tr}\left((u_j v_j^*)^* (u_i v_i^*) \right) = \operatorname{tr}\left(v_j u_j^* u_i v_i^* \right).$$

Because the left singular vectors are orthonormal, $u_j^* u_i = 0$ for $i \neq j$, so the whole trace is 0. Hence the rank-1 SVD components form an **orthonormal basis** (up to the σ_k scaling) for the subspace they span, in the Frobenius/matrix inner product. (Inner-product spaces for matrices are developed formally in L23.)

Connection to PCA

The SVD is essentially **Principal Component Analysis (PCA)** applied to a **data matrix**. Suppose you observe data vectors $x_1, \ldots, x_n$ and form the **sample correlation matrix**

$$R = \frac{1}{n} \sum_{i=1}^{n} x_i x_i^* = \frac{1}{n} X X^*, \qquad X = [x_1 \;\; \cdots \;\; x_n] \text{ (data/snapshot matrix)}.$$

PCA looks at the **eigenvalue decomposition** of R and asks which eigenvalues (directions) are most significant. But computing the **SVD of the data matrix** X **directly** yields the **same** eigenvectors: the left singular vectors of X are the eigenvectors of $R = \frac{1}{n}XX^*$. So **PCA = SVD on the data matrix** (rather than eigendecomposition of the correlation matrix). This is the standard, numerically preferable route.

How Singular Values/Vectors Are Computed (Preview)

The singular values and **left/right singular vectors** come from the **eigenvalue decomposition of** A^*A **or** AA^* (detailed in L22): $\sigma_k = \sqrt{\lambda_k(A^*A)}$, with v_k the eigenvectors of A^*A and u_k of AA^*. There are efficient algorithms; the main conceptual route is via A^*A.

Formal Proof of the SVD (by Induced 2-Norm and Block Diagonalization)

This is the centerpiece of the lecture: an existence proof modeled on the **Schur decomposition** proof (extend a vector to an orthonormal basis, get a block form, recurse). The recalled Schur result was that every square matrix is unitarily triangularizable; here the same extension idea is used to get block diagonal pieces for SVD.

Setup

Let $\sigma_1 = \|A\|_{2,2}$, the induced 2-norm — the **maximum Euclidean gain**, which is the longest principal semi-axis from the geometric picture. By definition of the induced norm, there exist **unit vectors** v_1 (input) and u_1 (output) with

$$Av_1 = \sigma_1 u_1, \qquad \|v_1\|_2 = \|u_1\|_2 = 1.$$

Extend to Orthonormal Bases

As in the Schur proof: extend u_1 to an orthonormal basis $u_1, u_2', \ldots, u_m'$ of the output space (forming unitary U_1), and extend v_1 to an orthonormal basis $v_1, v_2', \ldots, v_n'$ of the input space (forming unitary V_1). (The primed vectors are **not** the final singular vectors; they merely complete the bases.)

Form $S_1 = U_1^* A V_1$

Compute $S_1 = U_1^* A V_1$. The first column: $Av_1 = \sigma_1 u_1$, and

$$u_1^*(\sigma_1 u_1) = \sigma_1 \text{ (since } u_1^* u_1 = 1), \qquad (u_k')^*(\sigma_1 u_1) = 0 \text{ (orthogonality)}.$$

So the first column of S_1 is $(\sigma_1, 0, \ldots, 0)^\top$. The top row has entries $w_k^* = u_1^* A v_k'$ (call this row vector w^*). Thus

$$S_1 = \begin{bmatrix} \sigma_1 & w^* \\ 0 & B \end{bmatrix},$$

a **block upper-triangular-looking** form. The claim to prove: $w = 0$, which makes S_1 block **diagonal** and lets the proof recurse.

The Key Trick: $w = 0$ from Unitary Invariance of the 2-Norm

The instructor recalls the relevant unitary-invariance facts: the Euclidean vector 2-norm is unitarily invariant, the Frobenius norm is unitarily invariant for matrices, and the induced 2-2 norm is also unitarily invariant. Since $S_1 = U_1^* A V_1$ is A sandwiched by unitaries, S_1 has the **same** 2-norm as A:

$$\|S_1\|_{2,2} = \|A\|_{2,2} = \sigma_1.$$

So for **every** unit vector x, $\|S_1 x\|_2 \le \sigma_1$. Choose the **clever** unit vector built from the first row of S_1:

$$x = \frac{1}{\sqrt{\sigma_1^2 + \|w\|_2^2}} \begin{bmatrix} \sigma_1 \\ w \end{bmatrix}$$

(its norm is 1 because the denominator is exactly the norm of $[\sigma_1;\, w]$). Then

$$S_1 x = \frac{1}{\sqrt{\sigma_1^2 + \|w\|^2}} \begin{bmatrix} \sigma_1^2 + \|w\|^2 \\ Bw \end{bmatrix},$$

so its squared norm is

$$\|S_1 x\|_2^2 = \frac{(\sigma_1^2 + \|w\|^2)^2 + \|Bw\|^2}{\sigma_1^2 + \|w\|^2} = (\sigma_1^2 + \|w\|^2) + \frac{\|Bw\|^2}{\sigma_1^2 + \|w\|^2}.$$

This must be $\le \sigma_1^2$. The second term is **non-negative**, so already

$$\sigma_1^2 + \|w\|^2 \le \|S_1 x\|_2^2 \le \sigma_1^2 \quad \Longrightarrow \quad \|w\|^2 \le 0.$$

A norm cannot be negative, so $\|w\|^2 = 0$, i.e. $w = 0$. Therefore

$$S_1 = \begin{bmatrix} \sigma_1 & 0 \\ 0 & B \end{bmatrix}$$

is **block diagonal**.

Recurse

Now apply the identical argument to the smaller $(m-1) \times (n-1)$ block B: its induced 2-norm is σ_2 ($\le \sigma_1$), giving a unit u_2, v_2, another orthonormal extension, and another block diagonalization $\begin{bmatrix} \sigma_2 & 0 \\ 0 & C \end{bmatrix}$. Continuing peels off $\sigma_1 \ge \sigma_2 \ge \cdots$ down the diagonal, accumulating unitary factors, and **proves the SVD** $A = U\Sigma V^*$. (The ordering $\sigma_1 \ge \sigma_2 \ge \cdots$ follows because each step takes the 2-norm of a submatrix of the previous.) This also confirms the geometric claim that the **pre-images v_k are orthonormal**.

SVD vs. Eigenvalue Decomposition (Summary)

Property	Eigenvalue Decomposition	SVD
Form	$A = T\Lambda T^{-1}$	$A = U\Sigma V^*$
Applies to	square only	**any** (rectangular) matrix
Bases	**one** basis (same for in/out)	**two** orthonormal bases
Side matrices	invertible T (unitary iff normal)	unitary U, unitary V
Diagonal entries	complex eigenvalues	real **non-negative** singular values
Always exists?	**no** (not all diagonalizable)	**yes** (always)

Eigendecomposition diagonalizes using the **same** basis (when possible); for **normal** matrices that basis is unitary. The SVD uses **different** orthonormal bases for input and output, works for **any** matrix, and **always exists** — which is why it is so powerful.

Instructor Remarks and Study Guidance

- The conceptual leap to SVD is **dropping the same-basis restriction** of similarity: two orthonormal bases (one in, one out) diagonalize **any** matrix.
- Geometrically, A maps the unit sphere to an **ellipsoid**; σ_k = semi-axis lengths, u_k = axis directions (output), v_k = pre-images (input), with $Av_k = \sigma_k u_k$. Degenerate (rank-deficient) cases give flattened ellipsoids.
- The **rank** is the number of nonzero singular values; the **four fundamental subspaces** read directly off the columns of U and V.
- The rank-1 components $\sigma_k u_k v_k^*$ are **orthogonal in the matrix inner product** $\langle A, B\rangle = \mathrm{tr}(B^*A)$ (proof via $u_i^* u_j = 0$).
- **PCA = SVD of the data matrix** (equivalently eigendecomposition of the correlation matrix).
- The **existence proof**: $\sigma_1 = \|A\|_{2,2}$ gives $Av_1 = \sigma_1 u_1$; extend to orthonormal bases; unitary invariance of the 2-norm + a clever unit vector forces the off-diagonal row $w = 0$, giving a block-diagonal S_1; recurse. Modeled on the Schur proof.
- Singular values come from the eigenvalues of A^*A (or AA^*); detailed in L22. The instructor says there are efficient algorithms, but the lecture's main conceptual route is through these eigenvalue decompositions.
- Next lecture will cover uses of SVD and additional matrix norms formed by combining the singular values/principal semi-axis lengths.
- The starting point of SVD is the **2-2 norm** — the instructor notes he has not seen an analogous construction starting from the 1-1 or ∞-∞ norm (an open curiosity).

Source and Coverage Note

Source: `corrected/lecture21_corrected.md`.

Coverage: Matrix-norm recap motivating SVD, including vectorization-based matrix norms and the Frobenius example; algebraic view ($A = U\Sigma V^*$, shapes, real nonneg diagonal, V^* convention); connection to and departure from similarity (drop the same-basis restriction → two orthonormal bases diagonalize any matrix); geometric view (sphere → ellipsoid, line segment as degenerate ellipsoid, rank-deficient and tall/fat degenerate cases, hand-drawn figure remark, ellipsoid-as-quadratic-level-set student Q&A, preimage-existence check, assigning σ_k, u_k, v_k with $Av_k = \sigma_k u_k$); reduced and full SVD with $AV = U\Sigma$, multiplication by V^*, extension to unitary matrices, arbitrary completion columns, null-space zero action, and the four fundamental subspaces table; outer-product/rank-1 sum form with multiplication steps as an A-adaptive orthonormal expansion; matrix inner product $\langle A, B\rangle = \mathrm{tr}(B^*A)$ and proof that the rank-1 components are orthogonal; PCA connection (SVD of data matrix = eigendecomposition of correlation matrix); preview of computing singular values via A^*A/AA^* and efficient algorithms; full formal existence proof (induced 2-norm σ_1, orthonormal-basis extension à la Schur, Schur triangularization recall, $S_1 = U_1^* A V_1$ block form, unitary-invariance facts + clever-unit-vector argument forcing $w = 0$, recursion); SVD-vs-eigendecomposition comparison table; next-lecture remarks on SVD uses and norms built from singular values; instructor curiosity about whether analogous constructions exist for 1-1 or ∞-∞ norms.